WSQ COURSEWARE

Agentic AI for Email Marketing Campaign

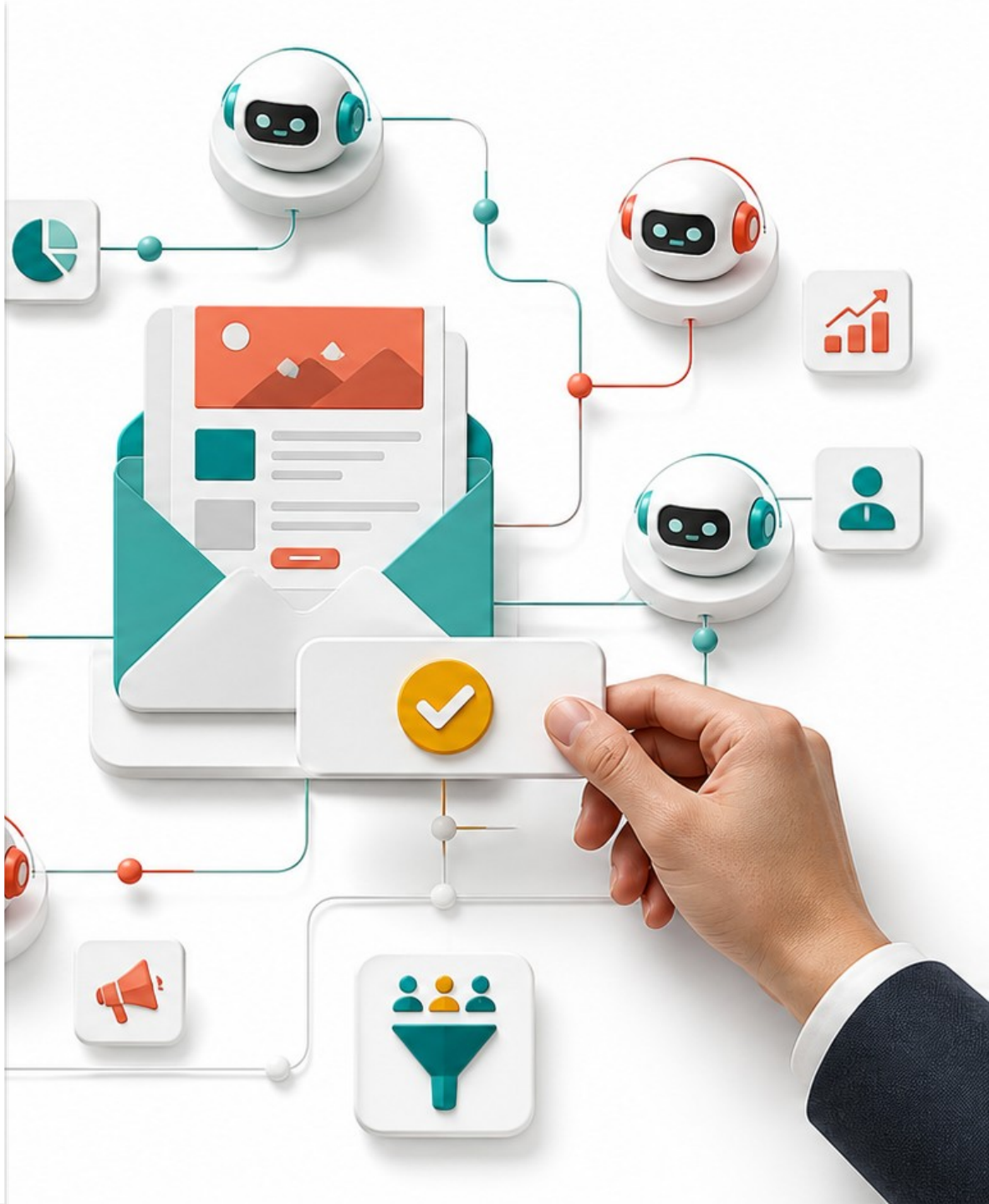
n8n · multi-agent editorial control · MailerLite

WSQ Course Code: TGS-2026064473

Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v1.1 · 24 August 2026

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved.



COURSE ADMINISTRATION

Welcome and Learning Contract

How the course runs, what evidence is required and where control boundaries sit

© 2026 Tertiary Infotech Academy Pte Ltd

Digital Attendance (Mandatory)

1 AM

Scan the SSG QR code at the start of each training day.

2 PM

Complete the second attendance event after lunch.

3 Assessment

Complete the separate assessment attendance event before the papers.

© 2026 Tertiary Infotech Academy Pte Ltd

TRAINER PROFILE

About the Trainer



Your Trainer

General trainer profile template

NAME _____

DESIGNATION _____

QUALIFICATIONS _____

EXPERTISE _____

© 2026 Tertiary Infotech Academy Pte Ltd

TRAINER PROFILE

About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE	Principal trainer and courseware developer
EXPERTISE	Agentic AI, automation, cloud and secure workflow design
DELIVERY	Technical WSQ courses with evidence-led labs
CONTACT	enquiry@tertiaryinfotech.com

© 2026 Tertiary Infotech Academy Pte Ltd

ICEBREAKER

Let's Know Each Other

1 Role

Your role and the audience you communicate with

2 Current stack

Email platform, CRM or automation tools you use

3 Goal

One campaign process you want to improve safely

© 2026 Tertiary Infotech Academy Pte Ltd

LEARNING CONTRACT

Ground Rules

1 Participate

Ask, challenge and test claims with evidence

2 Protect data

Use sample leads; never expose credentials or customer PII

3 Respect controls

Do not schedule to a live audience without approval

4 Stay on time

Return promptly from breaks

5 Work independently

Assessment evidence must be your own

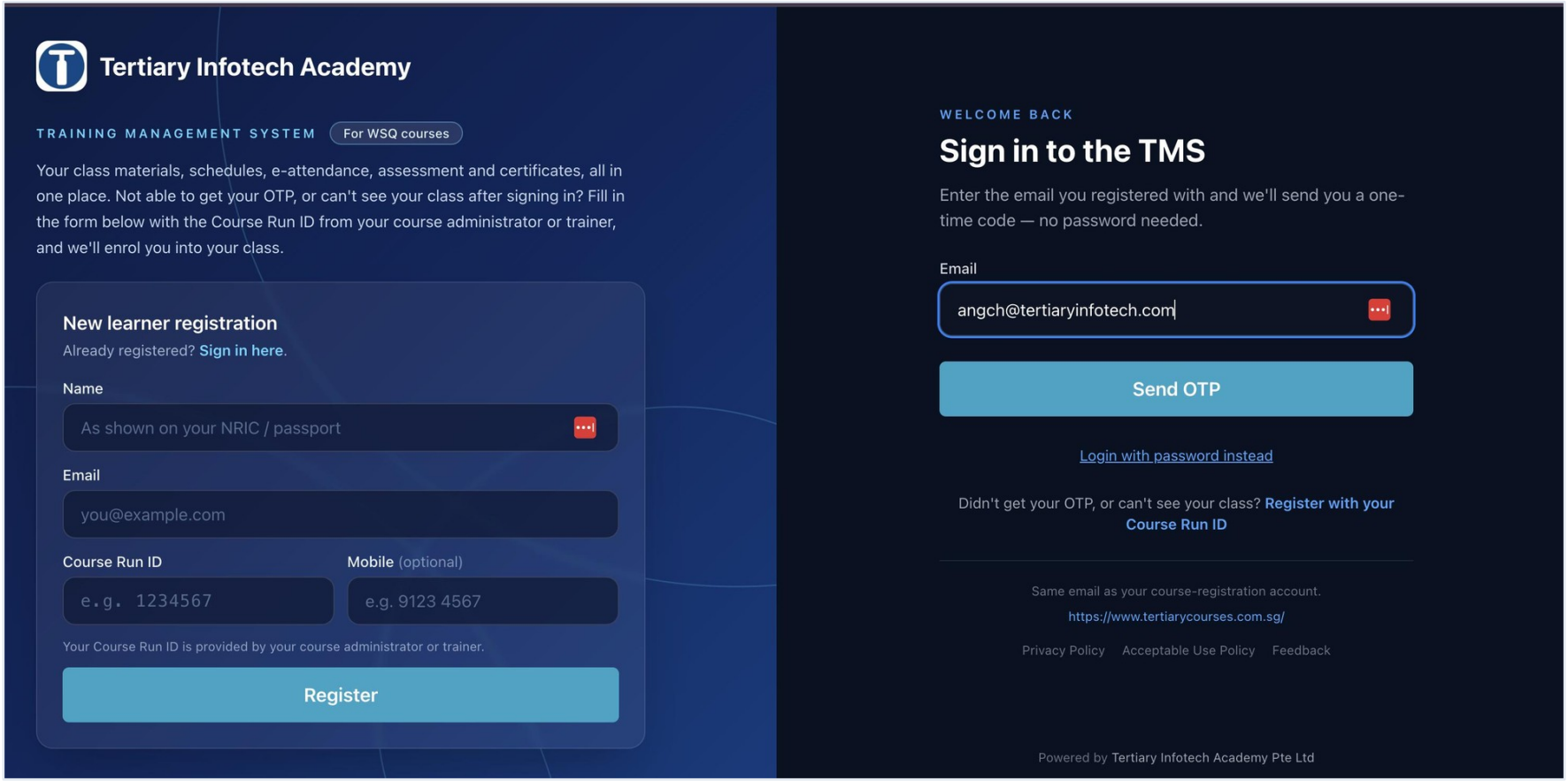
6 Attendance

At least 75% attendance is required

© 2026 Tertiary Infotech Academy Pte Ltd

COURSE PORTAL

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

© 2026 Tertiary Infotech Academy Pte Ltd

VISUAL PORTAL FLOW

Download Course Material

1

Sign in to lms-tms.tertiaryinfotech.com

2

Open this course run

3

Download slides, Learner Guide and lab package

4

Keep materials available for the open-book assessment

Read left to right. Each boundary has a named payload, state transition and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

SCHEDULE · 16 HOURS

Two-Day Learning Journey

1

Day 1 AM

System model, consent and campaign brief

2

Day 1 PM

Multi-agent editorial workflow and quality gate

3

Day 2 AM

Human approval, MailerLite draft and schedule

4

Day 2 PM

Analytics dashboard, trend/feedback optimization and assessment

© 2026 Tertiary Infotech Academy Pte Ltd

WSQ ALIGNMENT

Skills Framework

1

TSC title

E-Commerce Campaign Management

2

TSC code

TOU-SNM-3014-1.1-1

3

Course duration

2 days · 16 training hours

4

Assessment

WA 60 min + PP 60 min · open book

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT YOU WILL BE ABLE TO DO

Learning Outcomes

1 **LO1**

Implement campaigns aligned to strategic objectives

2 **LO2**

Analyse outcomes using sales data and consumer insights

3 **LO3**

Analyse market trends and customer reviews

© 2026 Tertiary Infotech Academy Pte Ltd

THREE ALIGNED TOPICS

Course Outline

1 Topic 1

Consent, campaign brief, multi-agent drafting, approval and MailerLite scheduling

2 Topic 2

Event facts, KPI model, cohort/funnel dashboard and diagnosis

3 Topic 3

Trend windows, feedback evidence, decision rules and experiments

© 2026 Tertiary Infotech Academy Pte Ltd

ASSESSMENT MAP

Knowledge and Ability Coverage

1 **K1-K2**

Architecture, roles, audience and message contract

2 **K3-K4**

Consent, subscriber lifecycle, approval and scheduling

3 **K5**

Metrics, event model and dashboard interpretation

4 **K6**

Trend, feedback and continuous improvement

5 **A1-A3**

Build the lead, multi-agent and release workflows

6 **A4-A5**

Build the dashboard and evidence-led optimization

© 2026 Tertiary Infotech Academy Pte Ltd

BEFORE ASSESSMENT

Briefing for Assessment

1 Materials

Open book: slides, Learner Guide and approved files only

2 Conduct

No discussion, photography or recording

3 Submission

Complete and upload on the LMS as instructed

4 Decision

All prescribed assessment requirements must be competent

© 2026 Tertiary Infotech Academy Pte Ltd

OPEN-BOOK ASSESSMENT

Assessment Instruments

WA · 60 minutes

- ✓ Six open-ended K-coded questions
- ✓ Knowledge from the deck and LG
- ✓ Answer every question

PP · 60 minutes

- ✓ Five connected A-coded tasks
- ✓ Evidence from the six labs
- ✓ Build, verify and interpret

© 2026 Tertiary Infotech Academy Pte Ltd

WA -> PP -> SIGN-OFF

Assessment Flow



Read left to right. Each boundary has a named payload, state transition and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

OPTIONAL PRACTICE

Practice Knowledge Portal

1 Portal

Open exams.tertiaryinfotech.com

2 Search

Use the portal search to find a relevant AI or automation practice set

3 Boundary

This WSQ course is assessed by the WA and PP papers, not an external certification exam

© 2026 Tertiary Infotech Academy Pte Ltd

TECHNICAL CORE

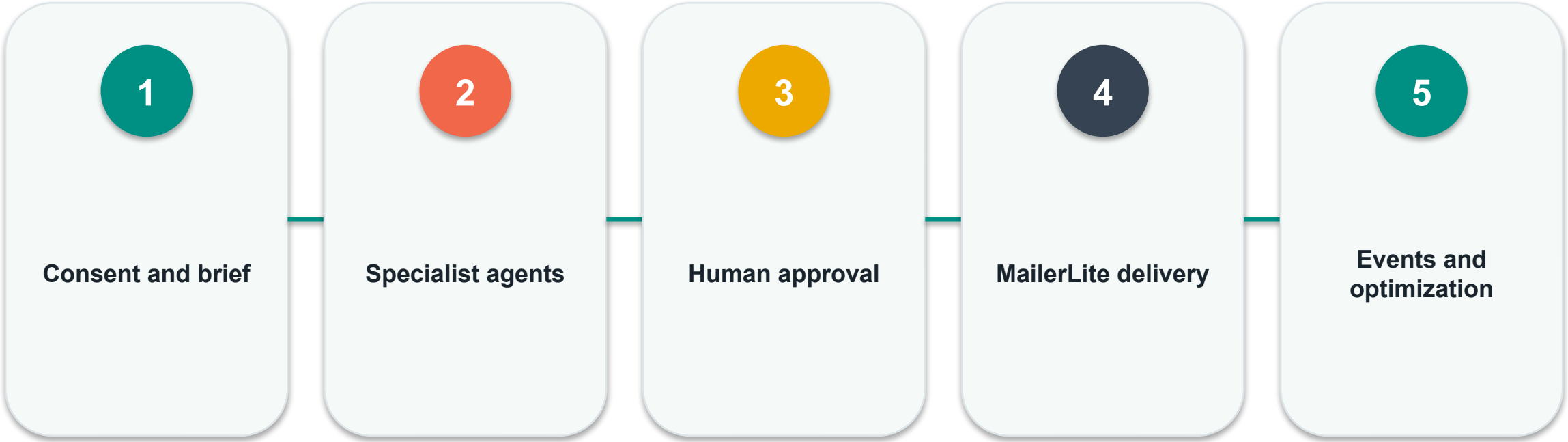
The Controlled Campaign Loop

Architecture, state, autonomy, evidence and failure controls

© 2026 Tertiary Infotech Academy Pte Ltd

SYSTEM MODEL

The Closed-Loop Campaign System



Read left to right. Each boundary has a named payload, state transition and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

DECISION MODEL

Automation, AI Workflow and Agentic System

Deterministic automation

- ✓ Fixed path and rules
- ✓ Predictable transforms
- ✓ Best for validation and writes

Agentic layer

- ✓ Chooses among bounded tools
- ✓ Handles ambiguous content work
- ✓ Must not own irreversible sends

© 2026 Tertiary Infotech Academy Pte Ltd

STATE MODEL

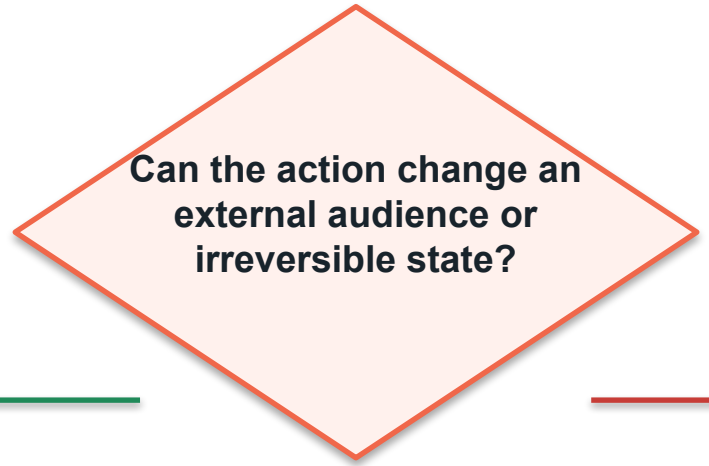
Campaign State Is a Contract

campaign_id	stable correlation key
status	draft | review | approved | scheduled | sent
version	monotonic content version
approval_hash	binds decision to exact payload
scheduled_at	ISO timestamp plus timezone
evidence_uri	audit and assessment artifact

© 2026 Tertiary Infotech Academy Pte Ltd

CONTROL

Autonomy Boundary for Email Marketing



YES · Require human review, payload preview, identity and expiry

NO · Allow bounded autonomous analysis with logging and limits

© 2026 Tertiary Infotech Academy Pte Ltd

WORKED MODEL

From Objective to Testable Campaign KPI

Target conversions = eligible subscribers x delivery rate x click rate x conversion rate

REALISTIC INPUTS

8,000 x 0.98 x 0.035 x 0.12 = 32.9 expected conversions

DECISION

A stronger subject line cannot repair a weak landing-page conversion rate.

© 2026 Tertiary Infotech Academy Pte Ltd

ARCHITECTURE

Three Data Planes

1 Control plane

brief, policy, approvals, schedule

2 Content plane

research, copy, HTML and variants

3 Evidence plane

events, metrics, logs and decisions

© 2026 Tertiary Infotech Academy Pte Ltd

EVIDENCE

Minimum Audit Record

1 **Evidence 1**

workflow_execution_id and campaign_id

2 **Evidence 2**

input brief version and source hashes

3 **Evidence 3**

agent outputs plus validation results

4 **Evidence 4**

reviewer decision, time, payload hash

5 **Evidence 5**

MailerLite response ID and final scheduled time

6 **Evidence 6**

error path, retry count and recovery action

© 2026 Tertiary Infotech Academy Pte Ltd

FAILURE CONTROL

Failure Taxonomy Before the Build

Content	hallucinated claim, tone drift, broken link
Audience	wrong group, stale consent, duplicate
Delivery	unverified sender, missing unsubscribe, timezone error
Measurement	open-rate bias, bot clicks, attribution gap
Operations	retry duplicates, leaked credential, unreviewed payload

© 2026 Tertiary Infotech Academy Pte Ltd

01

TOPIC 01

Implementing Agentic AI Email Marketing Campaigns

Consent -> brief -> specialist agents -> approval -> MailerLite schedule

© 2026 Tertiary Infotech Academy Pte Ltd

LO1 | K1-K4 | A1-A3

Topic 01 Coverage

1

Mechanism 1

Translate strategy into a machine-readable campaign brief and acceptance contract.

2

Mechanism 2

Separate audience, research, drafting and compliance roles across agent boundaries.

3

Mechanism 3

Gate every external send or schedule action with deterministic human approval.

4

Mechanism 4

Treat consent, unsubscribe state, secrets and audit evidence as first-class data.

© 2026 Tertiary Infotech Academy Pte Ltd

Specialists Propose; the Editor Decides



Audience agent
segment and job-to-be-done

Research agent
evidence and claims

Copy agent
subject, preheader and body

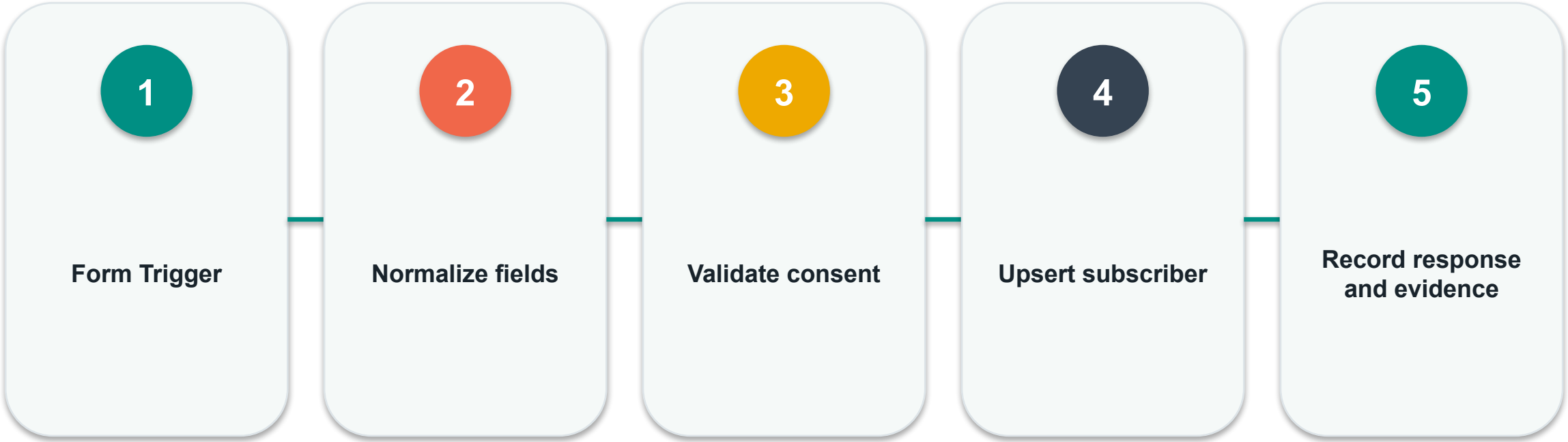
Compliance critic
policy and risk

Human editor
approve exact version

© 2026 Tertiary Infotech Academy Pte Ltd

N8N DATA FLOW

Lead Capture Runtime Trace



Read left to right. Each boundary has a named payload, state transition and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

CONCRETE CONTRACT

Consent Record Schema

email	normalized RFC-valid address
consent	explicit boolean; never inferred
consent_text_version	copy shown at submission
opted_in_at	UTC timestamp
optin_ip	where lawful and required
source	form or campaign attribution

© 2026 Tertiary Infotech Academy Pte Ltd

HTTP CONTRACT

MailerLite Subscriber Upsert

HTTP + JSON

```
POST /api/subscribers
Authorization: Bearer {{$credentials.apiKey}}
Content-Type: application/json

{"email":"{{$json.email}}","fields":{"name":"{{$json.name}}"},"groups":["{{$json.group_id}}"],"opted_in_at":"{{$json.opted_in_at}}"}
```

WHAT TO VERIFY 201 means created; 200 means the existing subscriber was updated non-destructively.

© 2026 Tertiary Infotech Academy Pte Ltd

SUBSCRIBER LIFECYCLE

Upsert Is Not Resubscribe

Safe upsert

- ✓ Adds or updates allowed fields
- ✓ Does not remove omitted groups
- ✓ Preserves status by default

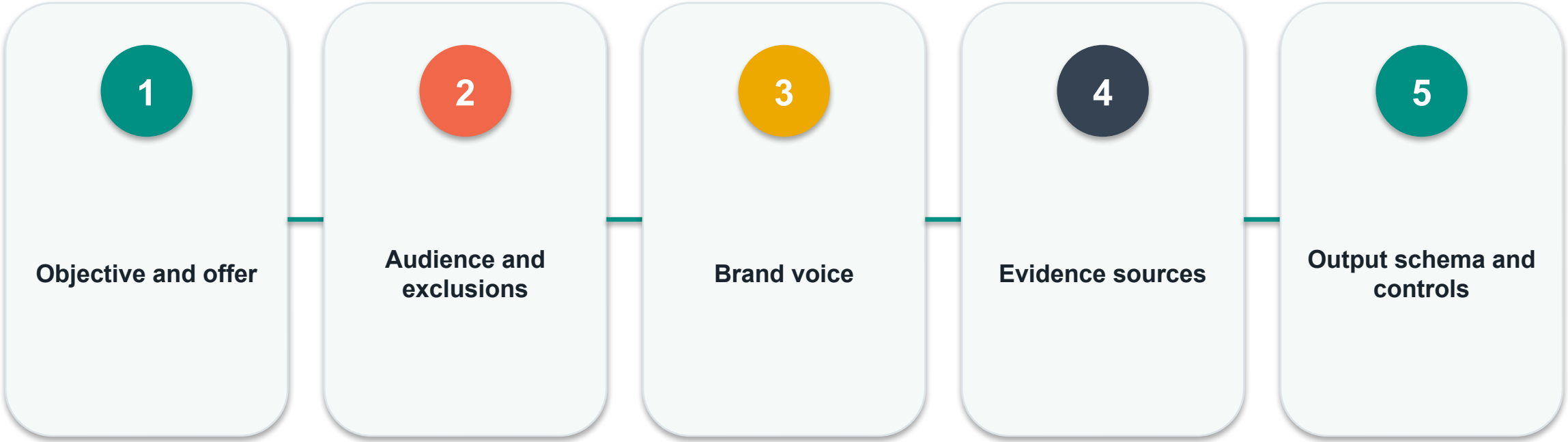
High-risk resubscribe

- ✓ Must respect prior unsubscribe
- ✓ Requires an authorised consent route
- ✓ Never set true by an agent guess

© 2026 Tertiary Infotech Academy Pte Ltd

CONTEXT ASSEMBLY

Campaign Brief Becomes Agent Context



Read left to right. Each boundary has a named payload, state transition and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

STRUCTURED OUTPUT

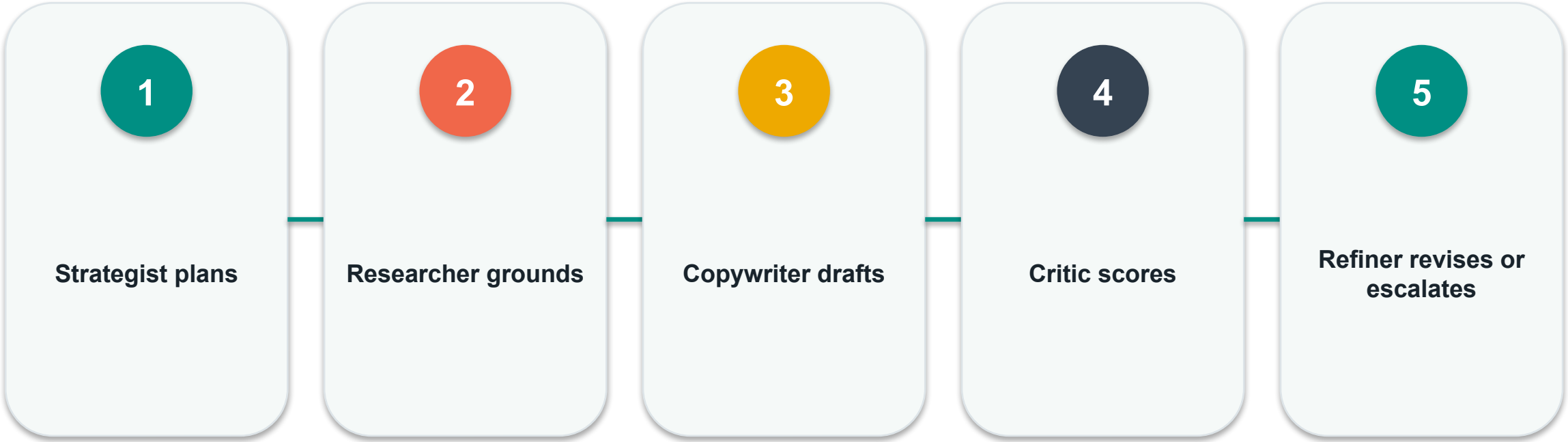
Newsletter Output Schema

subject	<= 60 characters, no unsupported claim
preheader	complements subject
html	valid semantic email HTML
text	plain-text alternative
cta	one primary action and tracked URL
citations	source IDs for factual claims
risk_flags	empty only after critic pass

© 2026 Tertiary Infotech Academy Pte Ltd

AGENT ORCHESTRATION

Multi-Agent Editorial Loop



Read left to right. Each boundary has a named payload, state transition and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

ORCHESTRATION CHOICE

Parallel Specialists vs Sequential Review

Parallel

- ✓ Faster independent viewpoints
- ✓ Requires merge contract
- ✓ Good for research and audience options

Sequential

- ✓ Each stage sees prior artifact
- ✓ Simpler provenance
- ✓ Good for draft -> critic -> revision

© 2026 Tertiary Infotech Academy Pte Ltd

DECISION RULE

Quality Gate as a Deterministic Score

Gate = min(brand, evidence, compliance, actionability)

REALISTIC INPUTS

Scores 92, 88, 74, 90 -> Gate 74 -> REVISE

DECISION

The weakest mandatory dimension controls release; averages can hide a compliance failure.

© 2026 Tertiary Infotech Academy Pte Ltd

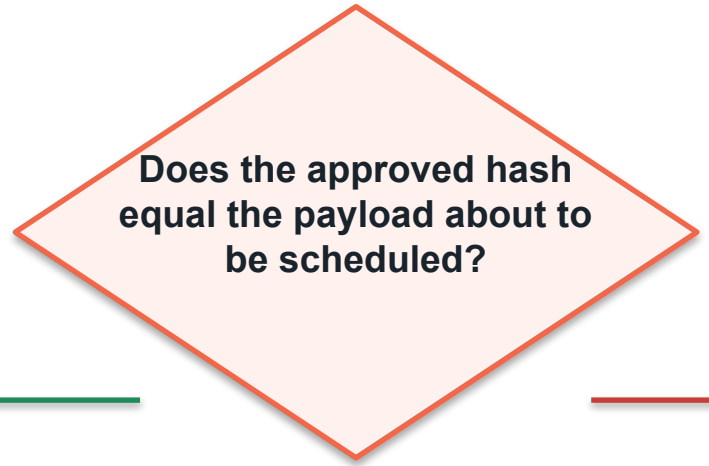
Reviewer Decision Payload

decision	approve | reject | revise
reviewer_id	authenticated reviewer
payload_hash	hash of exact draft and schedule
expires_at	approval time-to-live
feedback	required on reject or revise
version	must match pending draft

© 2026 Tertiary Infotech Academy Pte Ltd

TOCTOU CONTROL

Approval Must Bind to the Exact Payload



YES · Create or update the MailerLite draft, then schedule

NO · Block the write and return to review with a new version

© 2026 Tertiary Infotech Academy Pte Ltd

HTTP CONTRACT

Create a MailerLite Campaign Draft

HTTP + JSON

POST /api/campaigns

{"name":"{{$json.campaign_name}}","type":"regular","emails":
[{"subject":"{{$json.subject}}","from_name":"{{$json.from_name}}","from":"{{$json.verified_sender}}","content":"{{$json.html}}"}],"g
roups":["{{$json.group_id}}"]}

WHAT TO VERIFY The sender must already be verified. Advanced-plan rules may apply to HTML content.

© 2026 Tertiary Infotech Academy Pte Ltd

HTTP CONTRACT

Schedule Only After Approval

HTTP + JSON

POST /api/campaigns/{{$json.campaign_id}}/schedule

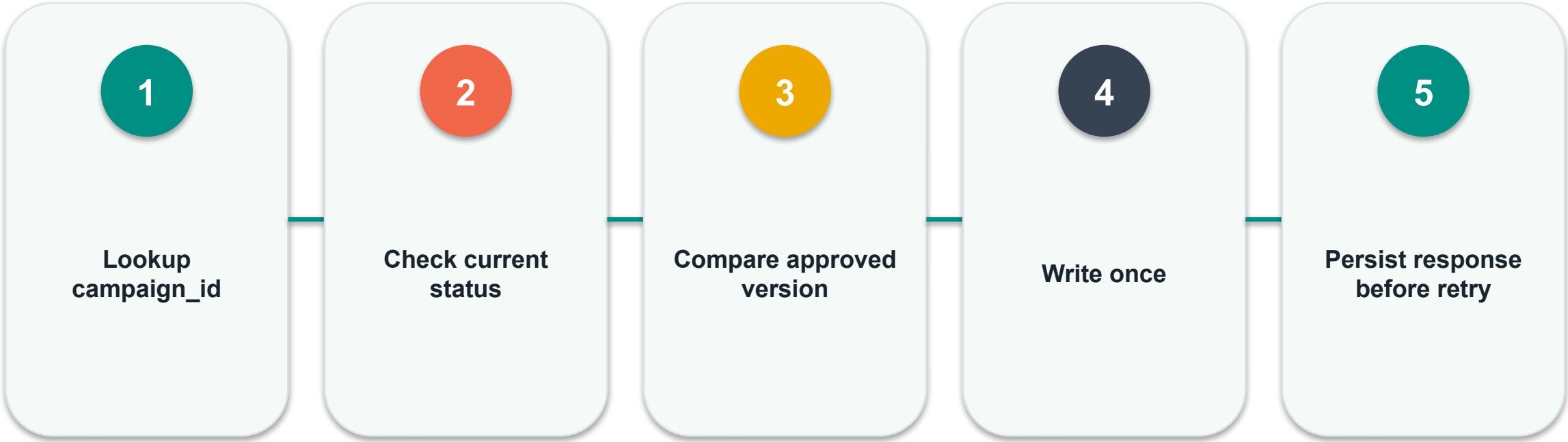
{"delivery":"scheduled","schedule":{"date":"2026-09-15","hours":"10","minutes":"00","timezone_id":{{$json.timezone_id}}}}

WHAT TO VERIFY Use a future date and preserve timezone explicitly. Store the returned campaign ID.

© 2026 Tertiary Infotech Academy Pte Ltd

RELIABILITY

Idempotent Schedule Pattern



Read left to right. Each boundary has a named payload, state transition and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

DIAGNOSTICS

Campaign-Write Failure Modes

401	credential or scope invalid
422	invalid sender, audience, content or schedule
429	rate limit; back off and retry safely
5xx	retry only with idempotency evidence
ready/sent	do not update as if still draft

© 2026 Tertiary Infotech Academy Pte Ltd

ASSESSMENT BRIDGE

LO1 Evidence Bundle

1 Evidence 1

lead-form test submission with explicit consent

2 Evidence 2

MailerLite 200/201 subscriber response

3 Evidence 3

multi-agent draft with critic score and revision

4 Evidence 4

review payload showing decision and hash

5 Evidence 5

MailerLite campaign ID and scheduled time

6 Evidence 6

n8n execution log with no secret values

© 2026 Tertiary Infotech Academy Pte Ltd

01

LAB 1

Consent-First Lead Capture to MailerLite

LO1 | K3 | A1 · 75 minutes

© 2026 Tertiary Infotech Academy Pte Ltd

PRACTICE BRIDGE

Lab 1 Scenario

1 Business need

Build an n8n form workflow that validates explicit consent, normalises a lead, upserts the subscriber into a MailerLite group and preserves audit evidence.

2 Deliverable

A production-shaped lead form, MailerLite upsert, rejection path and evidence record.

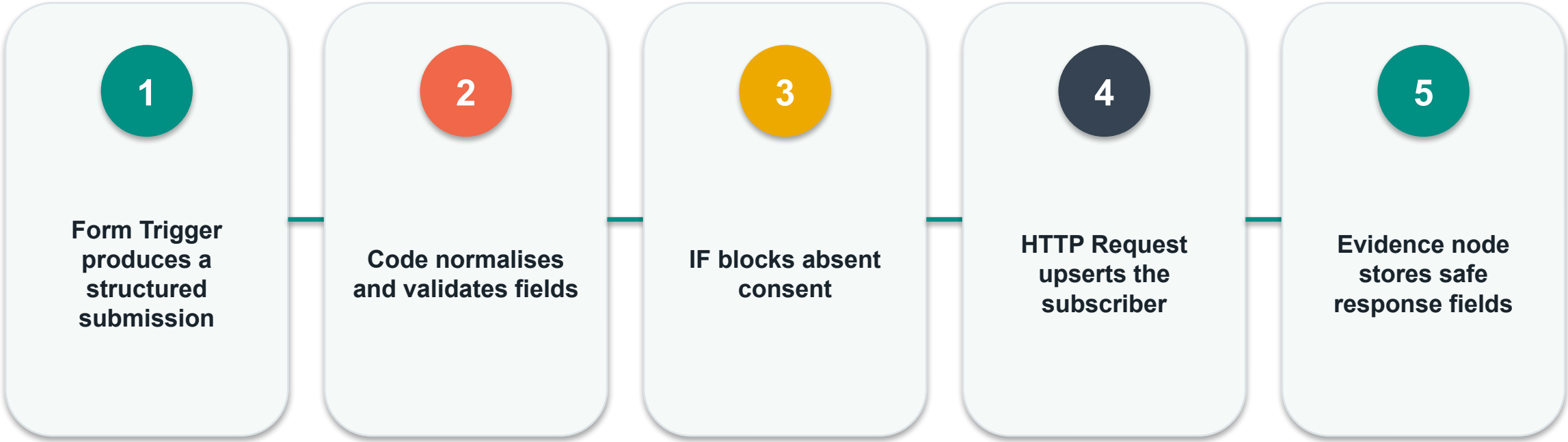
3 Tool boundary

n8n Form Trigger, Code, IF, HTTP Request, MailerLite API

© 2026 Tertiary Infotech Academy Pte Ltd

HOW THE WORKFLOW RUNS

Lab 1 Runtime Mechanism



Read left to right. Each boundary has a named payload, state transition and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT MUST REMAIN TRUE

Lab 1 Concrete Contract

Rule 1	email is trimmed and lowercased
Rule 2	consent must equal true
Rule 3	consent_text_version is recorded
Rule 4	group_id is configuration, not model output
Rule 5	API token remains in an n8n credential

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT TO CAPTURE

Lab 1 Evidence Package

1 **Evidence 1**

test submission

2 **Evidence 2**

normalised payload

3 **Evidence 3**

MailerLite subscriber ID and status

4 **Evidence 4**

n8n execution ID

© 2026 Tertiary Infotech Academy Pte Ltd

TEST WITHOUT CLICK PATHS

Lab 1 Verification and Failure

Accept

- ✓ valid consent reaches MailerLite
- ✓ missing consent never calls MailerLite
- ✓ subscriber ID/status captured
- ✓ credential value absent from exports

Diagnose

- ✓ invalid email returns a controlled form response
- ✓ unsubscribed status is never silently overridden
- ✓ 200 and 201 are both accepted
- ✓ 429 follows bounded exponential backoff

© 2026 Tertiary Infotech Academy Pte Ltd

02

LAB 2

Campaign Brief and Audience Contract

LO1 | K1-K2 · 60 minutes

© 2026 Tertiary Infotech Academy Pte Ltd

PRACTICE BRIDGE

Lab 2 Scenario

1 Business need

Turn a marketing objective into a validated campaign brief that every specialist agent consumes consistently.

2 Deliverable

A campaign-brief intake form, JSON schema validator and audience/message contract.

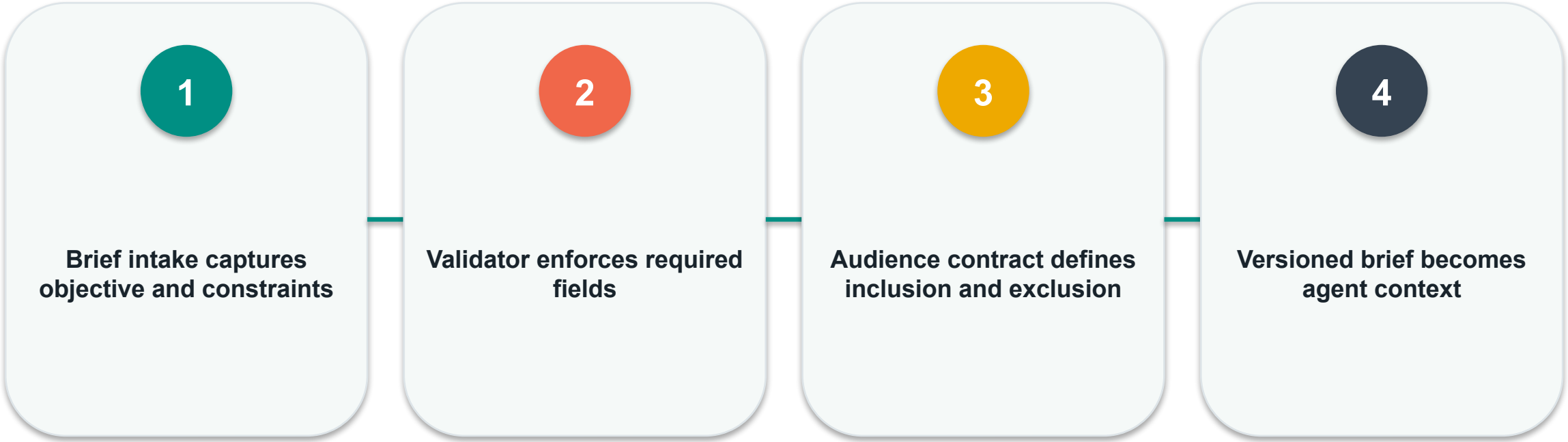
3 Tool boundary

n8n Form Trigger, Edit Fields, Code, Structured Output

© 2026 Tertiary Infotech Academy Pte Ltd

HOW THE WORKFLOW RUNS

Lab 2 Runtime Mechanism



Read left to right. Each boundary has a named payload, state transition and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT MUST REMAIN TRUE

Lab 2 Concrete Contract

Rule 1	one primary objective
Rule 2	one primary CTA
Rule 3	eligible group and exclusions
Rule 4	brand voice and prohibited claims
Rule 5	evidence source IDs
Rule 6	schedule and timezone

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT TO CAPTURE

Lab 2 Evidence Package

1 **Evidence 1**

campaign-brief.json

2 **Evidence 2**

validator result

3 **Evidence 3**

audience inclusion/exclusion table

4 **Evidence 4**

brief version hash

© 2026 Tertiary Infotech Academy Pte Ltd

TEST WITHOUT CLICK PATHS

Lab 2 Verification and Failure

Accept

- ✓ all required fields pass
- ✓ exclusions present
- ✓ one CTA only
- ✓ brief version and hash generated

Diagnose

- ✓ ambiguous objectives fail validation
- ✓ missing exclusions stop the pipeline
- ✓ unversioned edits create a new brief version
- ✓ PII is not inserted into prompts

© 2026 Tertiary Infotech Academy Pte Ltd

03

LAB 3

Multi-Agent Newsletter Editorial System

LO1 | K1-K2 | A2 · 120 minutes

© 2026 Tertiary Infotech Academy Pte Ltd

PRACTICE BRIDGE

Lab 3 Scenario

1 Business need

Coordinate strategist, researcher, copywriter, critic and refiner agents through explicit schemas and a bounded revision loop.

2 Deliverable

An importable multi-agent n8n workflow that produces a grounded newsletter draft with quality scores and provenance.

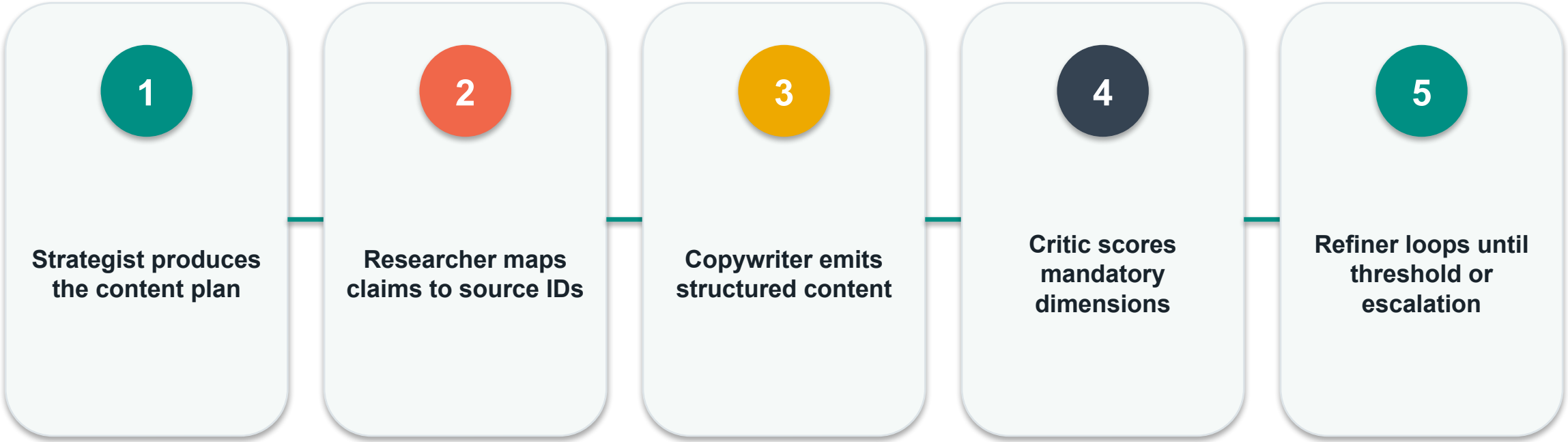
3 Tool boundary

n8n AI Agent, OpenAI Chat Model, Structured Output Parser, Loop Over Items, Code

© 2026 Tertiary Infotech Academy Pte Ltd

HOW THE WORKFLOW RUNS

Lab 3 Runtime Mechanism



Read left to right. Each boundary has a named payload, state transition and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT MUST REMAIN TRUE

Lab 3 Concrete Contract

Rule 1	structured JSON output
Rule 2	maximum two revisions
Rule 3	every factual claim carries a source ID
Rule 4	critic minimum score 80 on every dimension
Rule 5	no external send tool attached

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT TO CAPTURE

Lab 3 Evidence Package

1 **Evidence 1**

agent outputs

2 **Evidence 2**

critic scorecard

3 **Evidence 3**

revision count

4 **Evidence 4**

final content JSON

5 **Evidence 5**

execution graph

© 2026 Tertiary Infotech Academy Pte Ltd

TEST WITHOUT CLICK PATHS

Lab 3 Verification and Failure

Accept

- ✓ five distinct roles visible
- ✓ source IDs preserved
- ✓ schema valid
- ✓ bounded revision loop
- ✓ no send/schedule action

Diagnose

- ✓ parser errors route to repair
- ✓ missing source IDs force revision
- ✓ loop limit escalates to human
- ✓ agent outputs never overwrite raw brief

© 2026 Tertiary Infotech Academy Pte Ltd

04

LAB 4

Human Approval and MailerLite Scheduling

LO1 | K4 | A3 · 105 minutes

© 2026 Tertiary Infotech Academy Pte Ltd

PRACTICE BRIDGE

Lab 4 Scenario

1 Business need

Bind a human decision to the exact newsletter payload, create a MailerLite draft and schedule only the approved version.

2 Deliverable

A human-review gate, payload hash check, MailerLite draft creation and schedule transaction with release evidence.

3 Tool boundary

n8n Manual Trigger, Gmail Send and Wait, IF, Code, HTTP Request

© 2026 Tertiary Infotech Academy Pte Ltd

HOW THE WORKFLOW RUNS

Lab 4 Runtime Mechanism



Read left to right. Each boundary has a named payload, state transition and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT MUST REMAIN TRUE

Lab 4 Concrete Contract

Rule 1	authenticated reviewer
Rule 2	approval expiry
Rule 3	payload hash equality
Rule 4	verified sender
Rule 5	group ID allow-list
Rule 6	future schedule with timezone

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT TO CAPTURE

Lab 4 Evidence Package

1 **Evidence 1**

review request

2 **Evidence 2**

decision payload

3 **Evidence 3**

hash comparison

4 **Evidence 4**

MailerLite campaign ID

5 **Evidence 5**

scheduled timestamp

© 2026 Tertiary Infotech Academy Pte Ltd

TEST WITHOUT CLICK PATHS

Lab 4 Verification and Failure

Accept

- ✓ reject path makes no write
- ✓ hash mismatch blocks
- ✓ approved draft created
- ✓ campaign scheduled once
- ✓ evidence contains reviewer and campaign ID

Diagnose

- ✓ hash mismatch blocks write
- ✓ expired approval blocks release
- ✓ rejection records the decision without an external write
- ✓ 422 exposes safe validation detail

© 2026 Tertiary Infotech Academy Pte Ltd

02

TOPIC 02

AI-Driven Campaign Performance and Consumer Insights Analysis

MailerLite events -> normalised facts -> KPI model -> diagnostic dashboard

© 2026 Tertiary Infotech Academy Pte Ltd

LO2 | K5 | A4

Topic 02 Coverage

1

Mechanism 1

Model campaign, delivery and subscriber events at the correct grain.

2

Mechanism 2

Distinguish delivery, engagement and business outcomes in a metric tree.

3

Mechanism 3

Use cohort and funnel views to diagnose where performance is lost.

4

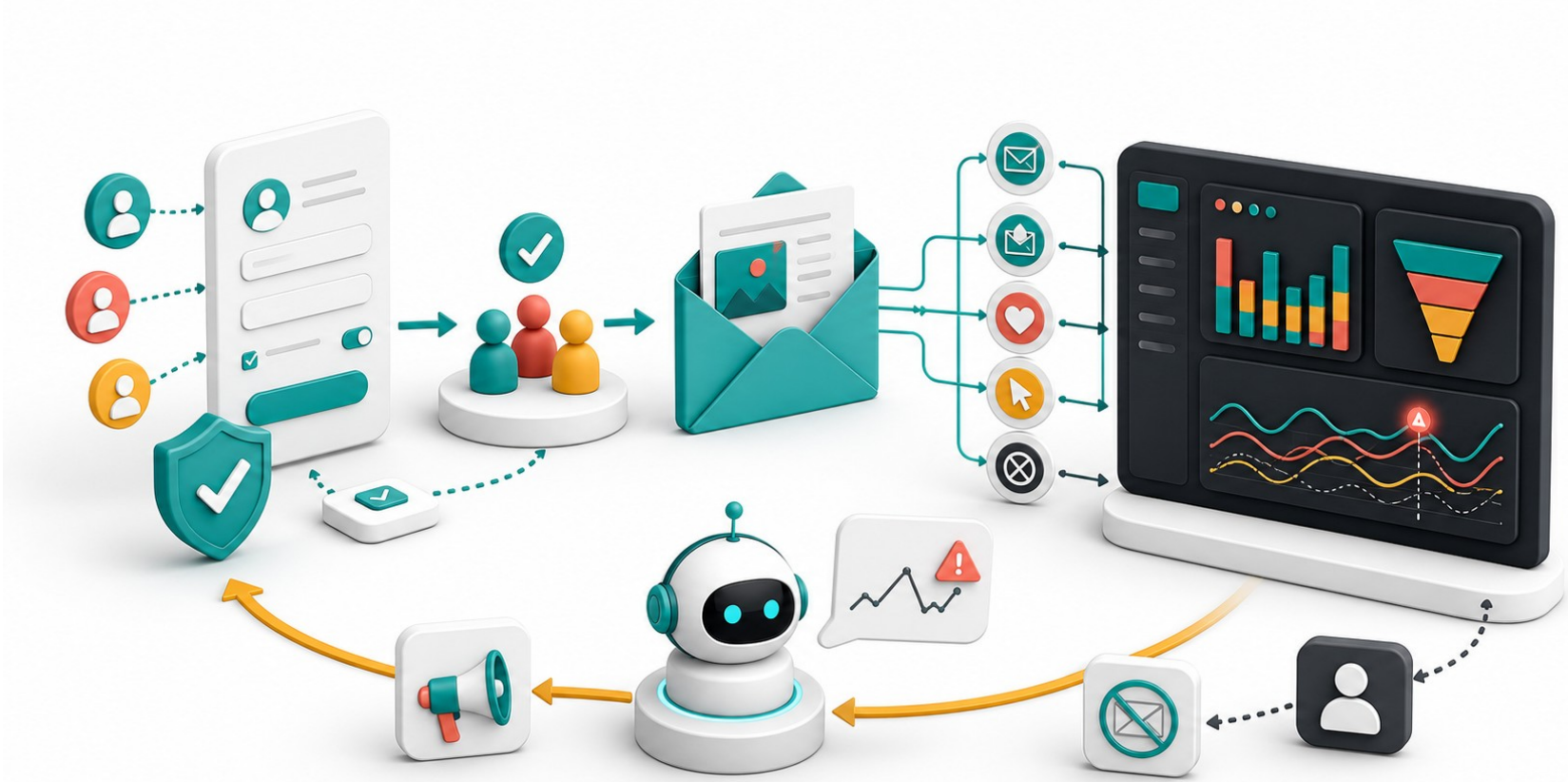
Mechanism 4

State limitations of open tracking, attribution and small samples.

© 2026 Tertiary Infotech Academy Pte Ltd

CLOSED LOOP

Acquisition, Delivery, Measurement, Optimization



Acquire
explicit consent

Deliver
approved campaign

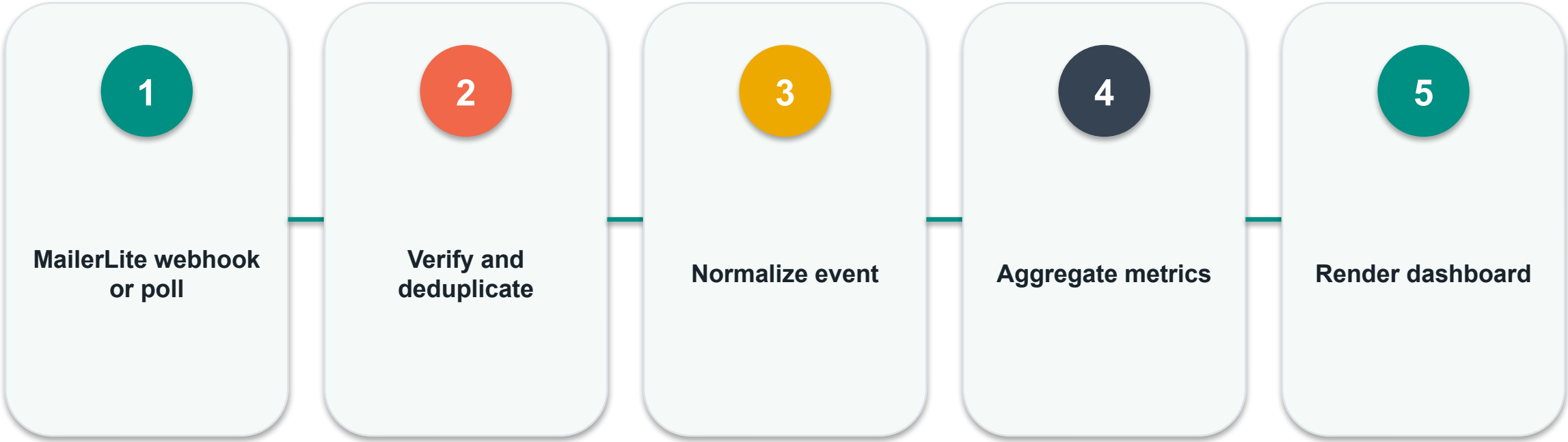
Measure
event facts

Optimize
controlled recommendation

© 2026 Tertiary Infotech Academy Pte Ltd

N8N DATA FLOW

Analytics Ingestion Runtime Trace



Read left to right. Each boundary has a named payload, state transition and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

DATA CONTRACT

Campaign Event Fact

event_id	deduplication key
event_type	sent | open | click | unsubscribe | bounce
occurred_at	event timestamp UTC
campaign_id	delivery identity
subscriber_id	pseudonymous analysis key
link_id	nullable click target
source_payload_hash	lineage without exposing PII

© 2026 Tertiary Infotech Academy Pte Ltd

SECURITY CONTRACT

Verify MailerLite Webhook Signature

JAVASCRIPT

```
const expected = hmacSha256(rawBody, webhookSecret);
if (!timingSafeEqual(signature, expected)) throw new Error('invalid signature');
return { eventId, eventType, occurredAt, campaignId };
```

WHAT TO VERIFY MailerLite signs webhook payloads with HMAC-SHA256. A valid 2xx response should be returned quickly.

© 2026 Tertiary Infotech Academy Pte Ltd

METRIC MODEL

Email KPI Definitions

Delivery rate = delivered / sent; CTR = unique clicks / delivered; CTOR = unique clicks / unique opens

REALISTIC INPUTS

9,600 delivered; 2,400 opens; 420 clicks -> CTR 4.38%, CTOR 17.5%

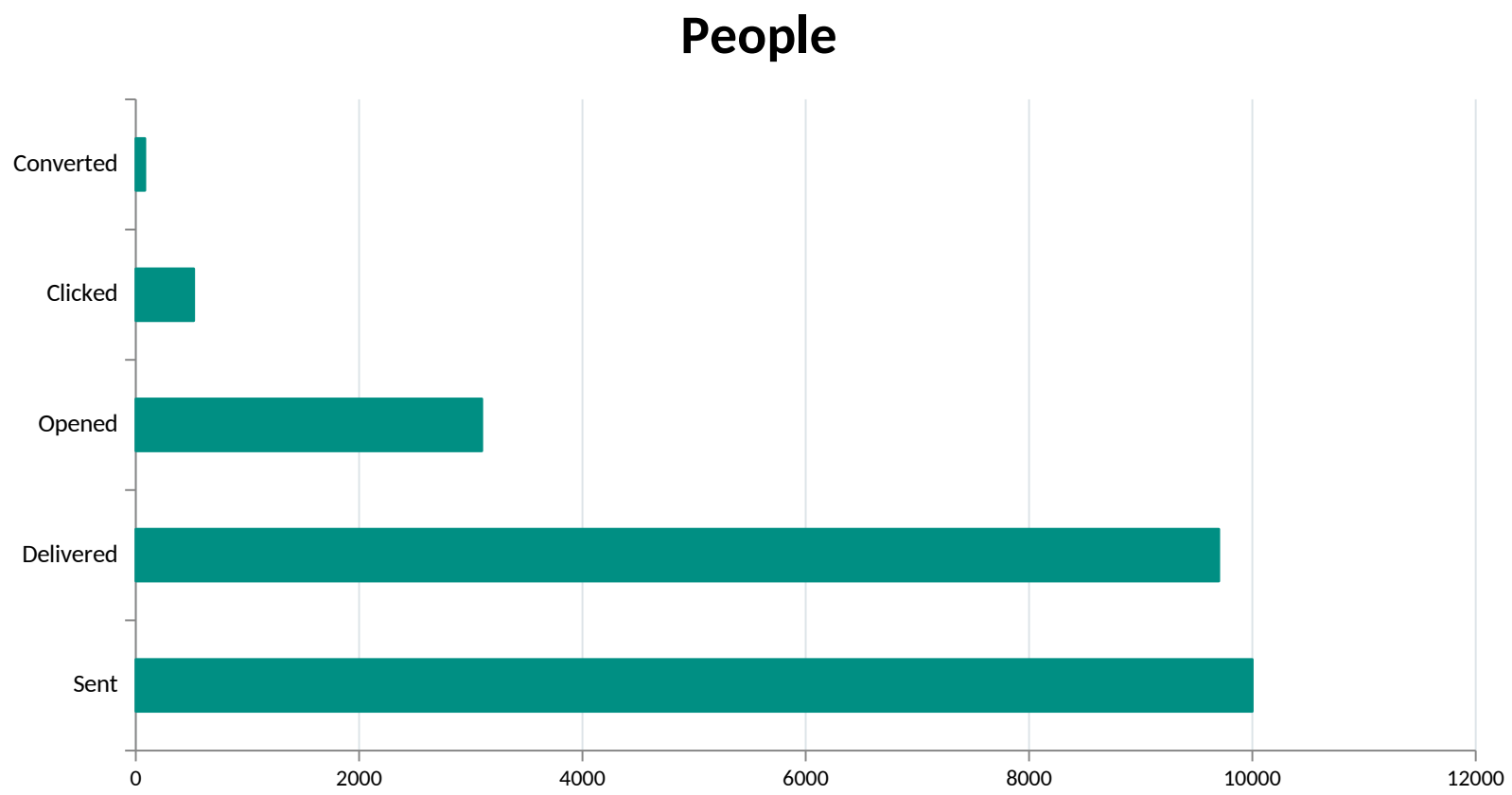
DECISION

Label the denominator on every chart; similarly named metrics can answer different questions.

© 2026 Tertiary Infotech Academy Pte Ltd

NATIVE PPT CHART

Funnel Reveals Where Value Is Lost



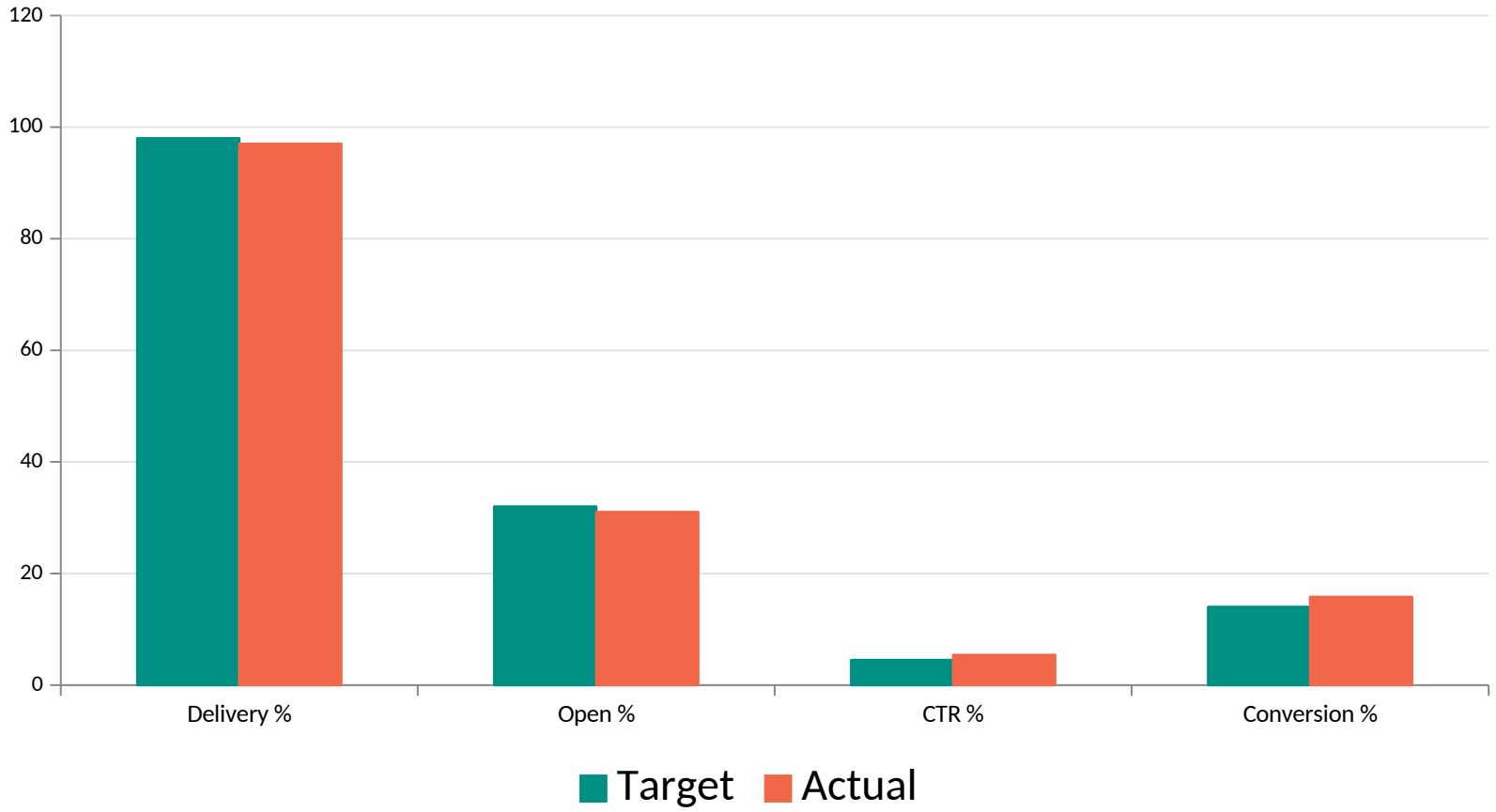
WHAT THE DATA SHOWS

The sharpest proportional drop is click-to-conversion, not delivery.

© 2026 Tertiary Infotech Academy Pte Ltd

NATIVE PPT CHART

Target vs Actual Campaign KPIs



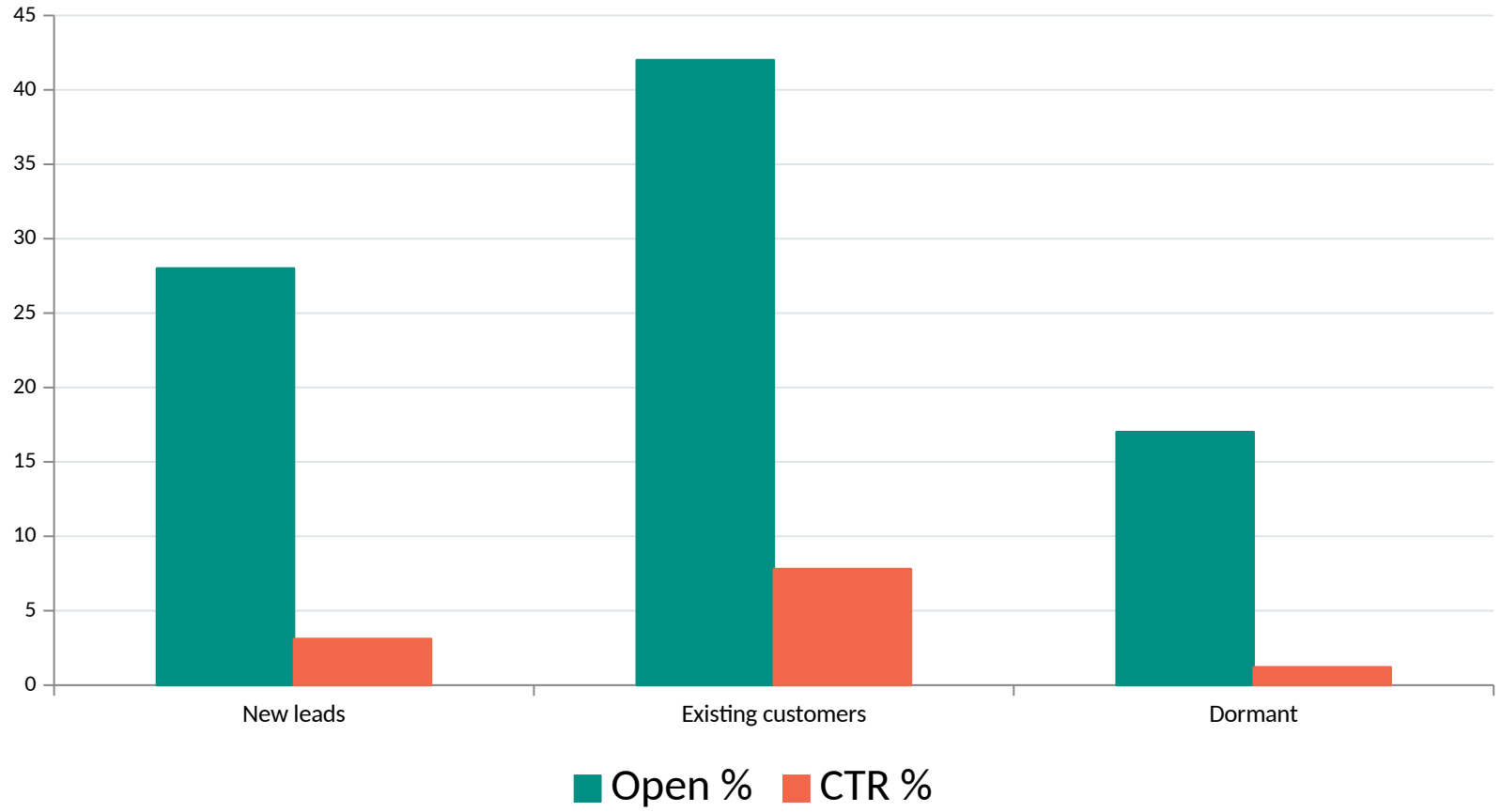
WHAT THE DATA SHOWS

Delivery and opens miss target; downstream engagement exceeds target.

© 2026 Tertiary Infotech Academy Pte Ltd

NATIVE PPT CHART

Cohort View Separates Audience Effects



WHAT THE DATA SHOWS

The aggregate hides a weak dormant cohort and a strong customer cohort.

© 2026 Tertiary Infotech Academy Pte Ltd

INGESTION CHOICE

Webhook vs Scheduled Poll

Webhook

- ✓ Near real-time events
- ✓ Requires HMAC verification
- ✓ Handle retries and duplicates

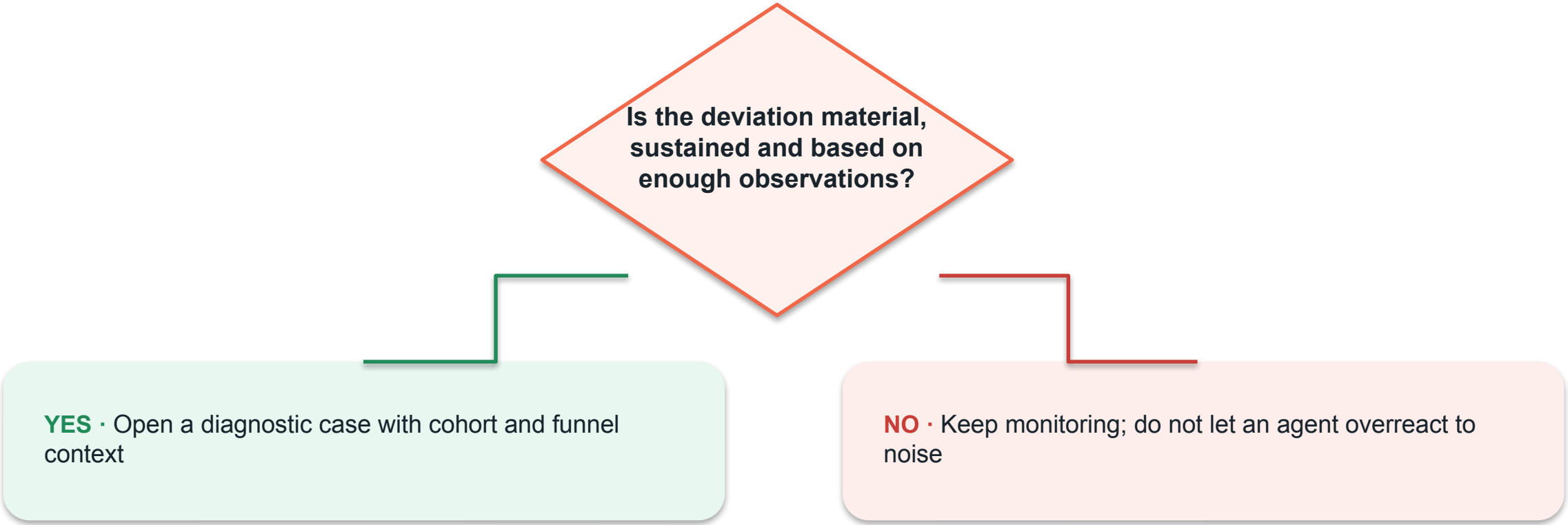
Scheduled poll

- ✓ Simpler recovery and backfill
- ✓ Higher latency and API use
- ✓ Use cursors and watermarks

© 2026 Tertiary Infotech Academy Pte Ltd

DIAGNOSTIC RULE

When Is an Alert Actionable?



© 2026 Tertiary Infotech Academy Pte Ltd

INTERPRETATION CONTROL

Measurement Limitations

Open tracking	privacy proxies and blocked pixels distort counts
Click activity	security scanners can create bot clicks
Attribution	last-click can ignore assisted conversions
Small cohorts	rates swing with few observations
Survivorship	unsubscribed users disappear from future engagement

© 2026 Tertiary Infotech Academy Pte Ltd

ASSESSMENT BRIDGE

LO2 Evidence Bundle

1 **Evidence 1**

validated event samples and deduplication result

2 **Evidence 2**

metric dictionary with denominators

3 **Evidence 3**

dashboard screenshot or HTML export

4 **Evidence 4**

target-versus-actual interpretation

5 **Evidence 5**

one diagnosed issue with evidence

6 **Evidence 6**

one controlled corrective action and expected metric movement

© 2026 Tertiary Infotech Academy Pte Ltd

05

LAB 5

MailerLite Campaign Analytics Dashboard

LO2 | K5 | A4 · 120 minutes

© 2026 Tertiary Infotech Academy Pte Ltd

PRACTICE BRIDGE

Lab 5 Scenario

1 Business need

Fetch current MailerLite campaign statistics, normalise the report, calculate labelled metrics and publish a self-contained diagnostic dashboard.

2 Deliverable

A live MailerLite campaign-report fetch, validated KPI calculation and HTML dashboard with status, target and funnel views.

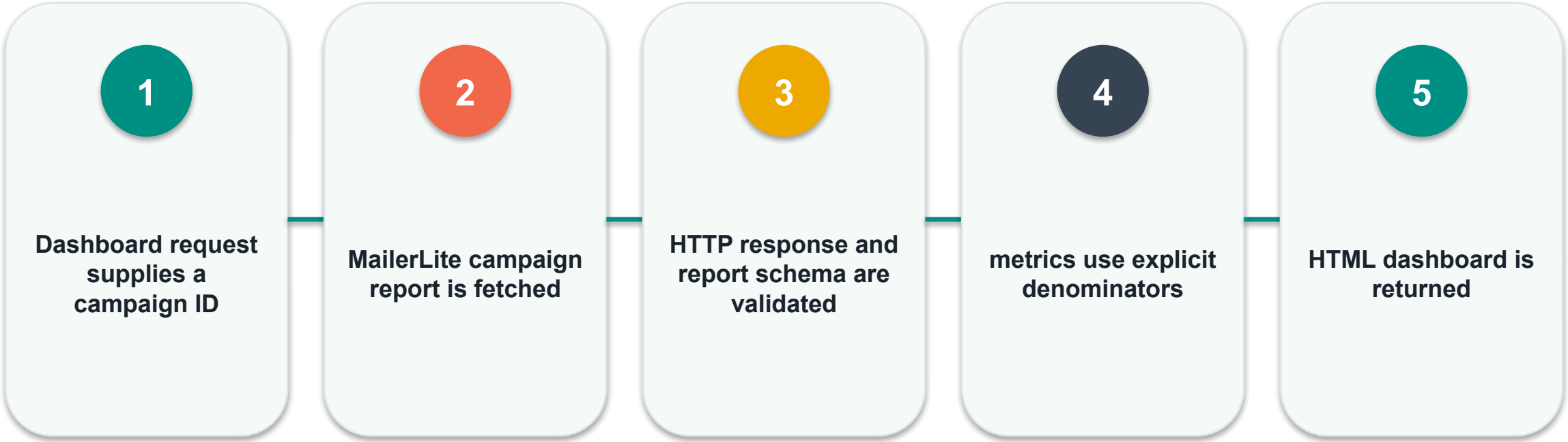
3 Tool boundary

n8n Webhook, Set, HTTP Request, Code, Respond to Webhook

© 2026 Tertiary Infotech Academy Pte Ltd

HOW THE WORKFLOW RUNS

Lab 5 Runtime Mechanism



Read left to right. Each boundary has a named payload, state transition and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT MUST REMAIN TRUE

Lab 5 Concrete Contract

Rule 1	campaign ID allow-listed
Rule 2	HTTP status validated
Rule 3	metric denominators labelled
Rule 4	zero denominators return N/A
Rule 5	PII excluded from dashboard

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT TO CAPTURE

Lab 5 Evidence Package

1 **Evidence 1**

MailerLite report response

2 **Evidence 2**

validated metric object

3 **Evidence 3**

metric dictionary

4 **Evidence 4**

dashboard HTML or screenshot

5 **Evidence 5**

diagnostic narrative

© 2026 Tertiary Infotech Academy Pte Ltd

TEST WITHOUT CLICK PATHS

Lab 5 Verification and Failure

Accept

- ✓ live report fetched
- ✓ response validated
- ✓ metrics recomputable
- ✓ dashboard renders
- ✓ diagnosis cites status/funnel evidence

Diagnose

- ✓ 401 identifies a credential problem
- ✓ 404 identifies an invalid campaign ID
- ✓ missing statistics fail schema validation
- ✓ empty denominator returns null not infinity

© 2026 Tertiary Infotech Academy Pte Ltd

03

TOPIC 03

AI Agent Analysis of Market Trends and Customer Feedback

Signals -> evidence -> themes -> confidence -> controlled optimization

© 2026 Tertiary Infotech Academy Pte Ltd

LO3 | K6 | A5

Topic 03 Coverage

1 Mechanism 1

Separate observed signals from agent inference and recommended action.

2 Mechanism 2

Combine quantitative trends with review themes and unsubscribe reasons.

3 Mechanism 3

Use confidence, materiality and reversible-test thresholds before changing a campaign.

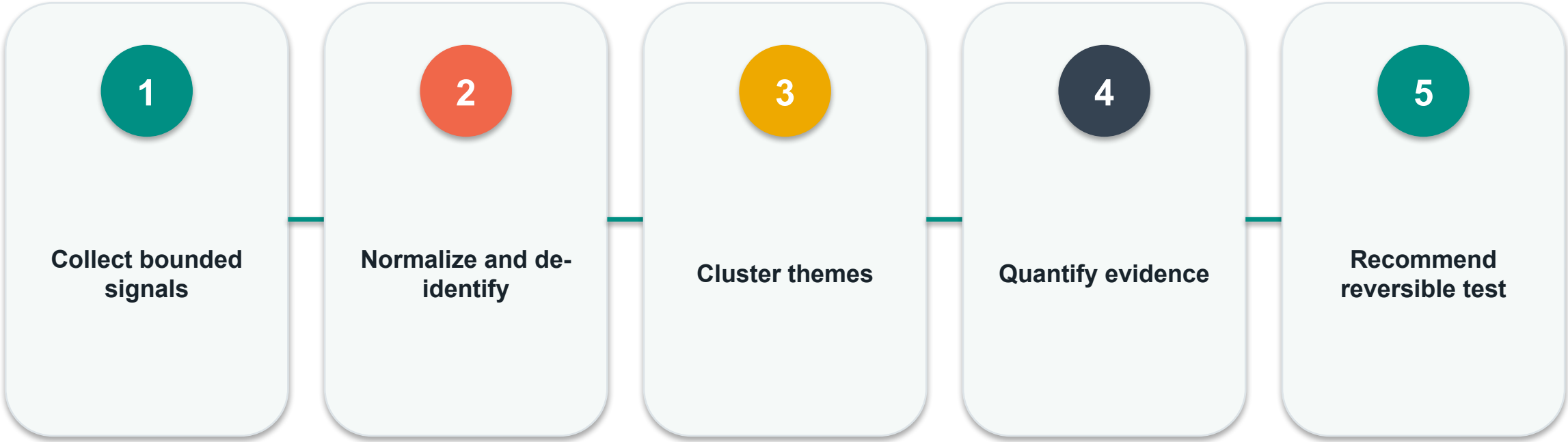
4 Mechanism 4

Write approved learnings back into the next campaign brief without contaminating raw data.

© 2026 Tertiary Infotech Academy Pte Ltd

AGENT ANALYSIS

Trend and Feedback Analysis Trace



Read left to right. Each boundary has a named payload, state transition and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

PROVENANCE CONTRACT

Evidence Card for Every Finding

finding	plain-language observation
source_ids	traceable records
sample_size	records behind the claim
window	analysis period
confidence	low | medium | high with reason
counter_evidence	what weakens the claim

© 2026 Tertiary Infotech Academy Pte Ltd

REASONING BOUNDARY

Signal, Inference and Recommendation

Observed

- CTR fell from 5.4% to 3.2%
- 27 unsubscribe comments mention frequency
- Dormant cohort had 1.2% CTR

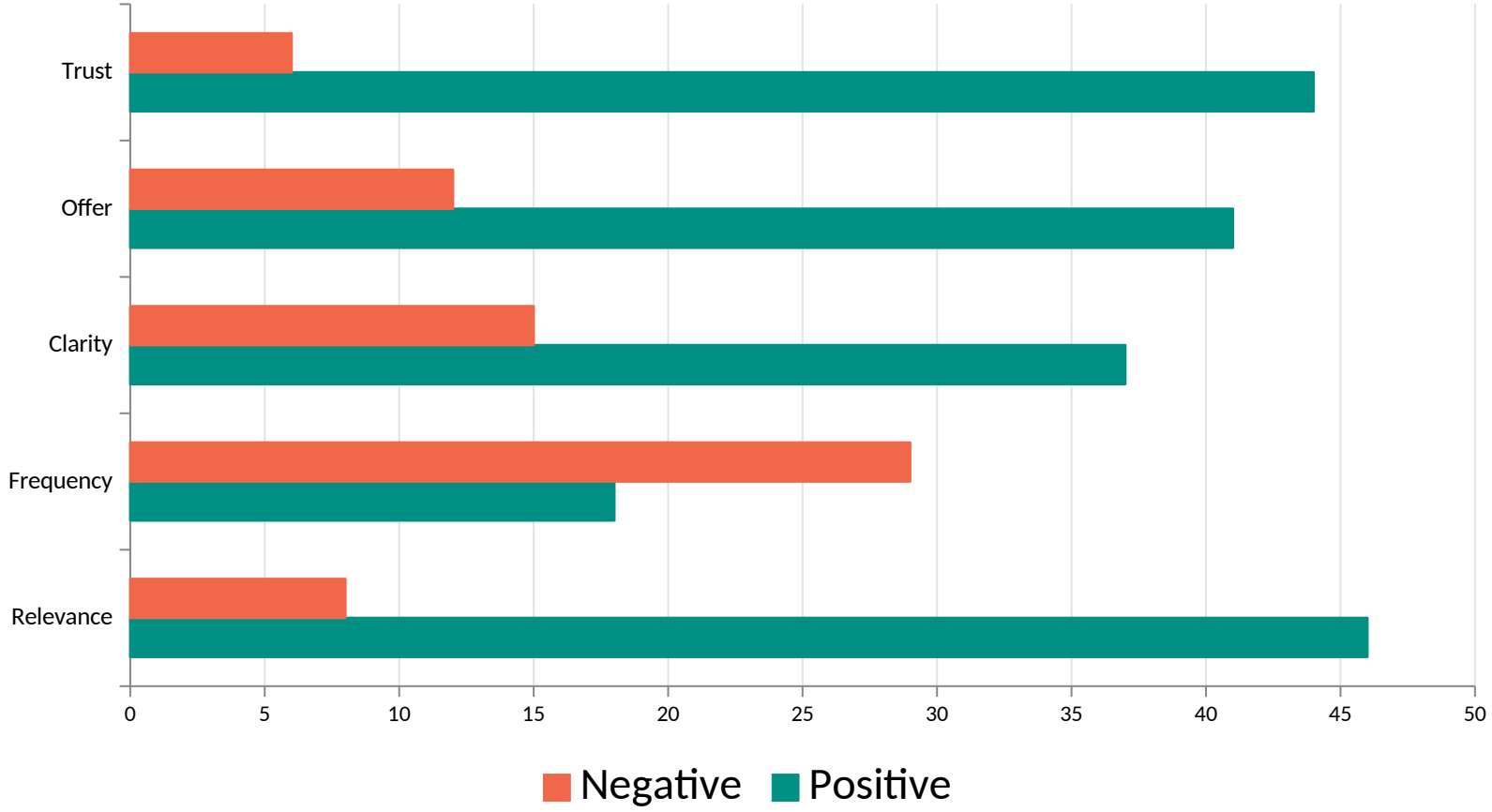
Inferred / proposed

- Frequency may be contributing
- Test a lower cadence
- Do not claim causality yet

© 2026 Tertiary Infotech Academy Pte Ltd

NATIVE PPT CHART

Feedback Themes by Campaign



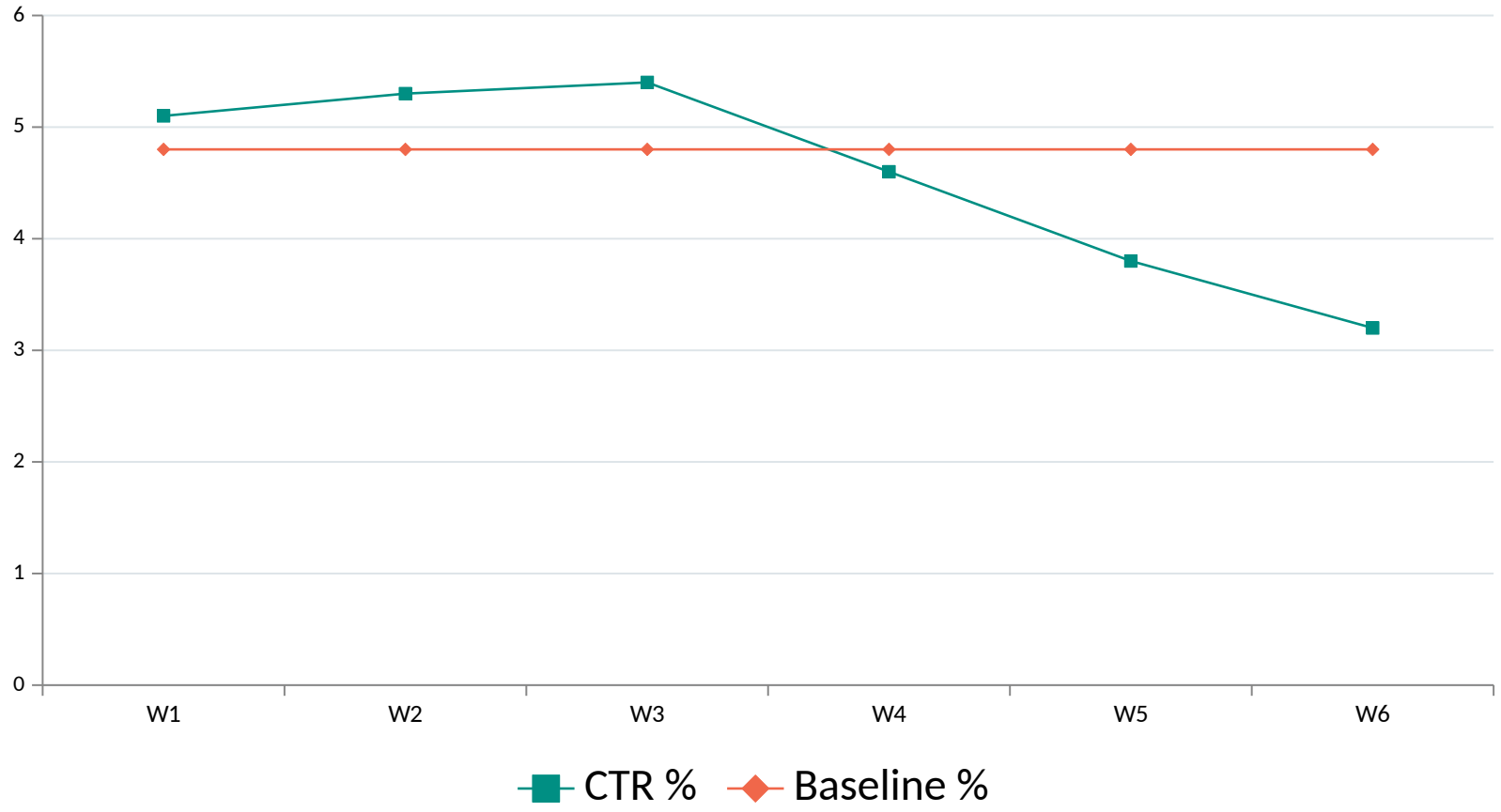
WHAT THE DATA SHOWS

Frequency is the only theme with more negative than positive mentions.

© 2026 Tertiary Infotech Academy Pte Ltd

NATIVE PPT CHART

Trend Needs a Window and Baseline



WHAT THE DATA SHOWS

Three consecutive weeks below baseline justify investigation, not an automatic send change.

© 2026 Tertiary Infotech Academy Pte Ltd

DECISION MODEL

Priority Score for Improvement Ideas

Priority = impact x confidence x reversibility / effort

REALISTIC INPUTS

Cadence test: 4 x 3 x 4 / 2 = 24; redesign: 4 x 2 x 2 / 5 = 3.2

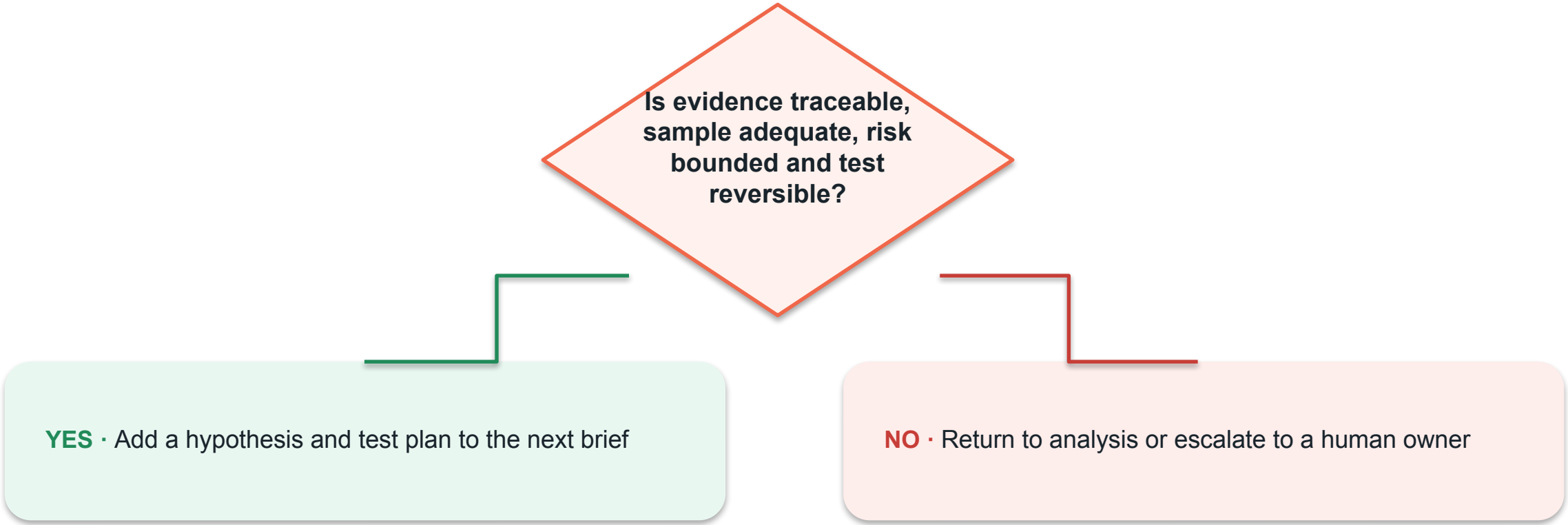
DECISION

The score ranks tests; it does not replace consent, compliance or brand review.

© 2026 Tertiary Infotech Academy Pte Ltd

CONTROL

Agent Recommendation Release Gate



© 2026 Tertiary Infotech Academy Pte Ltd

EXPERIMENT DESIGN

Optimization Experiment Contract

hypothesis	cause and expected direction
primary_metric	one decision metric
guardrail_metrics	unsubscribe, complaint, bounce
segment	eligible population
allocation	control and treatment split
stop_rule	time or sample threshold

© 2026 Tertiary Infotech Academy Pte Ltd

DIAGNOSTICS

Feedback-Analysis Failure Modes

PII leakage	raw reviews copied into prompts or slides
Theme collapse	distinct issues merged into one label
Sentiment shortcut	tone score replaces actual evidence
Recency bias	latest comments dominate
Action inflation	agent proposes change without baseline or owner

© 2026 Tertiary Infotech Academy Pte Ltd

ASSESSMENT BRIDGE

LO3 Evidence Bundle

1 **Evidence 1**

de-identified feedback data set

2 **Evidence 2**

theme counts linked to source IDs

3 **Evidence 3**

trend chart with window and baseline

4 **Evidence 4**

finding with counter-evidence and confidence

5 **Evidence 5**

ranked recommendation using decision rule

6 **Evidence 6**

experiment contract with guardrails and human owner

© 2026 Tertiary Infotech Academy Pte Ltd

06

LAB 6

Trend and Customer Feedback Optimization Agent

LO3 | K6 | A5 · 90 minutes

© 2026 Tertiary Infotech Academy Pte Ltd

PRACTICE BRIDGE

Lab 6 Scenario

1 Business need

Combine campaign trends with de-identified feedback, require evidence cards and produce a reversible experiment recommendation for human approval.

2 Deliverable

A trend/feedback analysis workflow with theme counts, confidence, counter-evidence, priority score and experiment contract.

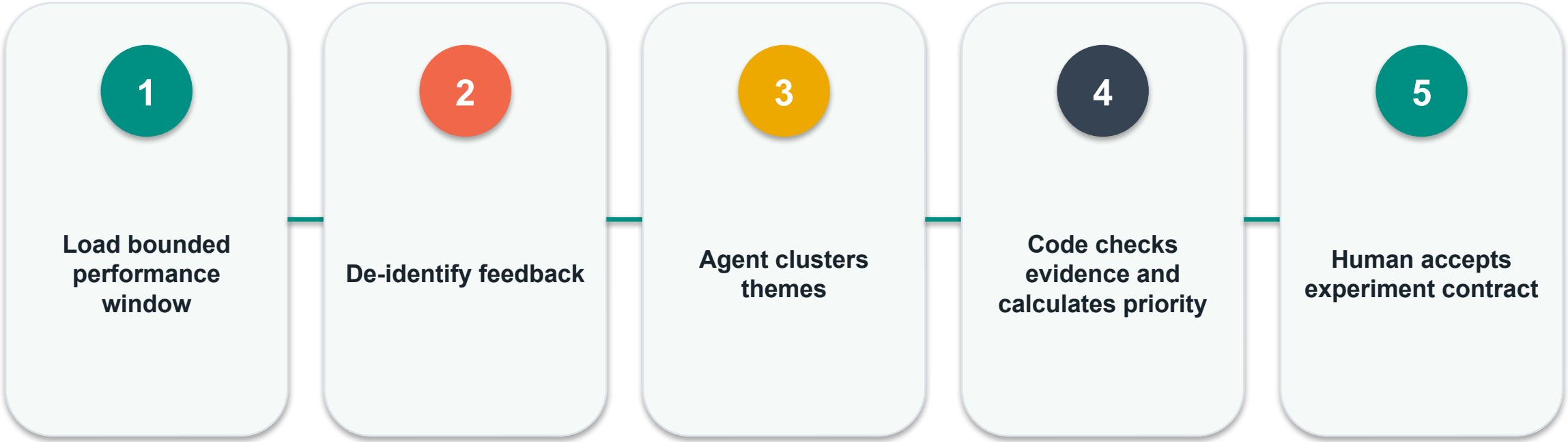
3 Tool boundary

n8n Code, AI Agent, Structured Output Parser, IF, Chat HITL Tool

© 2026 Tertiary Infotech Academy Pte Ltd

HOW THE WORKFLOW RUNS

Lab 6 Runtime Mechanism



Read left to right. Each boundary has a named payload, state transition and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT MUST REMAIN TRUE

Lab 6 Concrete Contract

Rule 1	no raw email addresses in prompts
Rule 2	theme counts link to source IDs
Rule 3	confidence includes reason
Rule 4	counter-evidence mandatory
Rule 5	experiment has guardrail metrics

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT TO CAPTURE

Lab 6 Evidence Package

1 **Evidence 1**

de-identified feedback

2 **Evidence 2**

theme table

3 **Evidence 3**

trend chart

4 **Evidence 4**

evidence card

5 **Evidence 5**

approved experiment contract

© 2026 Tertiary Infotech Academy Pte Ltd

TEST WITHOUT CLICK PATHS

Lab 6 Verification and Failure

Accept

- ✓ PII removed
- ✓ themes trace to sources
- ✓ trend has baseline/window
- ✓ counter-evidence present
- ✓ experiment has guardrails and approval

Diagnose

- ✓ insufficient sample returns monitor
- ✓ untraceable claim fails schema
- ✓ PII detector blocks prompt
- ✓ irreversible change requires escalation

© 2026 Tertiary Infotech Academy Pte Ltd

COURSE CLOSE

From Evidence to Assessment

Review the control loop, complete the assessment and preserve your working artifacts

© 2026 Tertiary Infotech Academy Pte Ltd

LO1 -> LO2 -> LO3

Revision Map

1 Implement

Labs 1-4: consent, brief, agents, approval and schedule

2 Analyse

Lab 5: event facts, metrics, cohorts and funnel

3 Improve

Lab 6: trends, feedback, evidence cards and experiments

© 2026 Tertiary Infotech Academy Pte Ltd

OPEN BOOK · 2 HOURS

Assessment Reminder

WA · Knowledge

- ✓ K1-K6 from the deck and LG
- ✓ Open-ended responses
- ✓ Use exact contracts and decision rules

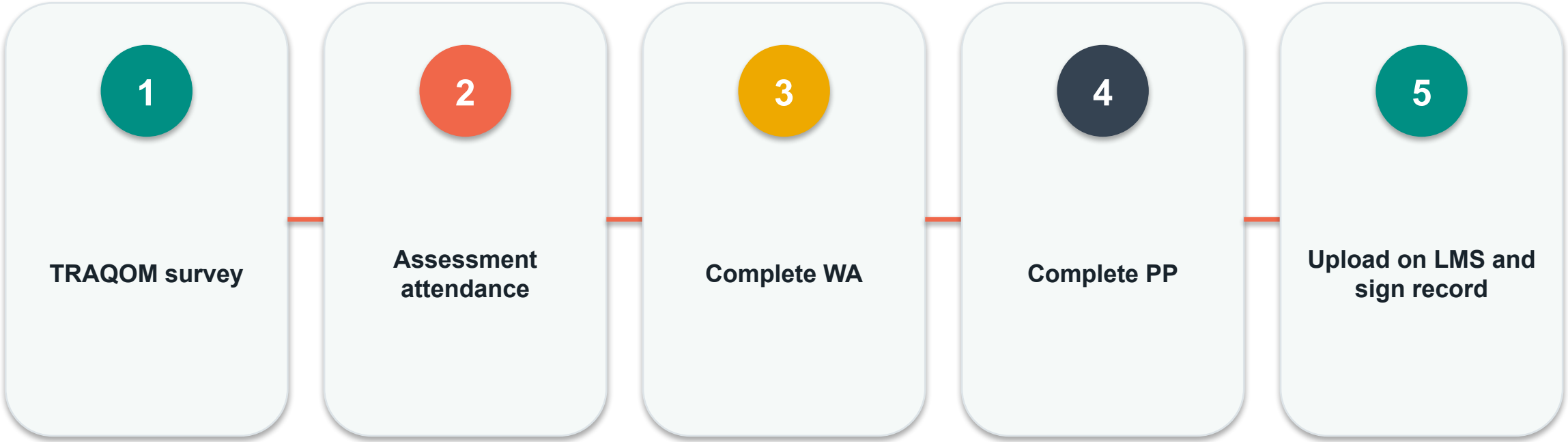
PP · Ability

- ✓ A1-A5 from the labs
- ✓ Show observable evidence
- ✓ Explain verification and failure handling

© 2026 Tertiary Infotech Academy Pte Ltd

FINAL CHECK

Assessment Flow



Read left to right. Each boundary has a named payload, state transition and owner.

© 2026 Tertiary Infotech Academy Pte Ltd

COURSE FEEDBACK

TRAQOM Survey (Mandatory)

1 Locate

Open the LMS course run

2 Scan

Use the displayed TRAQOM QR code

3 Submit

Complete the survey before final sign-off

© 2026 Tertiary Infotech Academy Pte Ltd

MANDATORY DETAILS

Certificate Delivery

1 Name

Confirm the spelling of your name

2 Email

Confirm the address used for delivery

3 Record

Complete the required LMS/TRAQOM details

© 2026 Tertiary Infotech Academy Pte Ltd

FINAL ATTENDANCE

Digital Attendance (Mandatory)

1 **AM**

Attendance recorded

2 **PM**

Attendance recorded

3 **Assessment**

Complete the final SSG attendance event

© 2026 Tertiary Infotech Academy Pte Ltd

POST-COURSE SUPPORT

Support

1 Email

enquiry@tertiaryinfotech.com

2 Telephone

+65 6100 0613

3 Course page

https://www.tertiarycourses.com.sg/casl-agentic-ai-for-email-marketing-campaign.html

© 2026 Tertiary Infotech Academy Pte Ltd

Thank You

Build agentic campaigns that are useful, measurable and controlled.

TGS-2026064473